

Vacation Planner App – Developer Specification Document

■ Project Overview

The Vacation Planner App is a comprehensive tool that helps users: 1. Plan a vacation by building a personalized schedule. 2. Enjoy the vacation with real-time itinerary management. 3. Reflect on their trip via an interactive report including ratings, photos, and stats.

■ Core Features

1. Pre-Vacation Planning

- Users input all desired activities manually or import from suggestions. - System builds a custom itinerary based on: - Location & distance between events - Time availability and preferences - Activity types (outdoors, relaxation, dining, etc.) - Option to manually drag-and-drop rearrange or auto-generate again.

2. Vacation Mode (During the Trip)

- Daily itinerary with time blocks and reminders. - Real-time adjustments based on: - Weather (API integration) - Location (GPS-aware) - Delays or skipped activities - Offline support: access schedule without connectivity.

3. Post-Vacation Report

- Visual summary of: - Activities completed (checklist format) - User ratings per activity (1-5 stars or emoji sliders) - Overall vacation score or mood summary - Embeds uploaded photos and captions. - Timeline or map view of the journey.

■ Gamified Add-On Feature: Scavenger Hunt Mode

- Optional toggle during planning. - Generates location-based or activity-based challenges: - 'Find a building with a red door' - 'Snap a group selfie at sunset' - 'Try a dish you've never had before' - Tracks progress with photo or checkbox proof.

■ Collaborative Planning

- Invite friends or family to co-plan. - Each participant can: - Add activities - Vote or comment on suggestions - Lock in shared time slots

■ Integrations

- Google Maps or Apple Maps for: - Directions - Travel time estimation - Nearby recommendations - Yelp / TripAdvisor APIs for: - Reviews - Photos - Pricing

■ Memories and Media

- Allow users to upload: - Photos - Short video clips - Notes or journal entries - Content is tied to specific activities and displayed in the report.

■ Budget Tracking

- Add estimated and actual cost to each activity. - Visual budget tracker: pie chart or progress bar. - Final report includes: - Total spent - Category breakdown (e.g., food, travel, lodging)

■ Feedback and Ratings

- Prompt users immediately after each activity to rate it. - Optional free-text field: 'What did you love/hate?' - Results feed into report and suggestion engine.

■ Offline Mode

- Cache full itinerary, activity info, and map snapshots. - Automatically syncs when online.

■ Technical Considerations

Frontend

- React Native (iOS + Android support) - Calendar & card-based UI components - LocalStorage/SQLite for offline caching - Push notification service integration (e.g. Firebase)

Backend

- Node.js or Python (FastAPI) - PostgreSQL or MongoDB for user and itinerary data - Cloud storage for media uploads (e.g., S3) - External API integrations: - OpenWeatherMap - Google Maps / Yelp / TripAdvisor

Authentication

- OAuth (Google, Apple) - Optional family group sync via invitation links

Security

- Encrypted token-based auth (JWT) - Secure storage of media + user content - GDPR & CCPA compliance for data export/deletion